The empirical recurrence for OEIS A183909, proved from an exact model

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Abstract

OEIS A183909 carries an empirical recurrence of order 7. It is true, and it is decidable rather than empirical. The entry's sequence is given exactly by an exact polynomial in n, from which a provably satisfies a MONIC linear recurrence of order $S = 7$, derived below and not assumed. The residual of the conjectured recurrence satisfies the same one, so whether it annihilates a is settled by a finite exact computation. No transfer matrix is involved and the count is not a walk.

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1 The conjecture

OEIS A183909 is "Number of nondecreasing arrangements of n+2 numbers in 0..6 with each number being the sum mod 7 of two others.". It has offset 1 and begins

$$1, 1, 22, 136, 470, 1193, 2525, 4752, \ \ldots$$

The entry states, as a conjecture contributed by its author and never marked settled:

> Empirical: a(n) = (1/720)*n^6 + (11/240)*n^5 + (89/144)*n^4 + (209/48)*n^3 - (13003/360)*n^2 + (531/10)*n + 12 for n>3. a(n) = 7*a(n-1) - 21*a(n-2) + 35*a(n-3) - 35*a(n-4) + 21*a(n-5) - 7*a(n-6) + a(n-7) for n>10.

As of the "Last modified" line on the live entry (Apr 06 2018 07:00:59, revision 8) this is still recorded as empirical, and nothing on the entry records it as proved.

2 The count is a polynomial

A NONDECREASING arrangement is determined by its multiset of values, so an admissible arrangement is exactly an admissible multiset. The condition names how many other entries a value must be built from, and a condition of that shape cannot distinguish a value occurring $r+1$ times from one occurring more: the admissible multisets are therefore exactly those whose multiplicity profile, CAPPED at $r+1$, is one of a finite list, and that list is computed once by enumeration over the capped profiles.

Given an admissible capped profile with F the total multiplicity of the values capped below the limit and t the number of values at the cap, the arrangements of length ℓ with that profile number $\binom{\ell-F-(r+1)t+t-1}{t-1}$, and $a(n)$ is the sum of those binomials over the list with ℓ the entry's own length.

3 The bound

Each binomial is a polynomial in ℓ, hence in n, of degree $t-1 < M$ with M the alphabet size, so a agrees with a polynomial of degree below M and $(z-1)^M$ annihilates it: $S = 7$.

Two kinds of term are outside that polynomial at small lengths, and they are why the threshold below is not zero. A profile with no value at the cap contributes the indicator $[F = \ell]$, which is a single spike rather than a polynomial and is gone once ℓ exceeds every such F; and a profile whose slack falls below $-(t-1)$ has the count 0 where the binomial polynomial does not vanish. Both live at bounded ℓ, so above them a is the polynomial exactly, and on every entry of this family the residual of $(z-1)^M$ is nonzero at the first few indices and vanishes from then on. The theorem is stated from the threshold up, and the run of zeros it uses lies entirely above the transient.

4 The proof

Lemma 1. *Let $A(z) = z^S - \sum_{j<S} \alpha_j z^j$ be the monic annihilator derived above and $q(z) = z^7 - \sum_i c_i z^{7-i}$ the characteristic polynomial of the conjectured recurrence. The residual $r(n) = a(n) - \sum_i c_i a(n-i)$ is a linear combination of shifts of a, so A annihilates r as well. Since $A(0) \neq 0$ the recurrence it gives runs in both directions, and S consecutive zeros of r force r to vanish at every later index.*

Proof. A annihilates a, hence every shift of a, hence any linear combination of shifts; that is r. A monic recurrence with nonzero constant term determines each term from the S before it and each from the S after it, so a run of S zeros propagates. ☐

Theorem 2.

$$a(n) = 7\,a(n-1) - 21\,a(n-2) + 35\,a(n-3) - 35\,a(n-4) + 21\,a(n-5) - 7\,a(n-6) + a(n-7)$$

for every $n > 10$.

Proof. The terms $a(0), \ldots$ were computed exactly in integer arithmetic from the model, extended by A where the model itself was not evaluated, and $r(n)$ formed for each index. Those integers vanish from $n = 10 + 1$ onwards, and the run exceeds $S + 7$, which by Lemma 1 settles every larger index at once. The bound is exact: at $n = 10$ the recurrence fails on the entry's own published terms, so no larger range holds. ☐

5 Verification

Three checks, each independent of the algebra above.

First, the model reproduces all 34 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry: it is built by reading the entry's English, and a misreading gives different counts.

Second, the conjectured recurrence was evaluated directly on those published terms, with no model involved, and holds wherever the proved range and the data overlap.

Third, the derived annihilator A was itself tested on terms it had not been given: the model was evaluated beyond the S values that determine it and A reproduced them. A bound that is too small fails this test, which is why it is run.

References

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[4] F. Diamond and J. Shurman, *A First Course in Modular Forms*, Springer GTM 228, 2005.